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| **Fiscal Year:** | FY 2005 | **Task Last Updated:** | FY 03/23/2006 |
| **PI Name:** | Cowings, Patricia S. Ph.D. | | |
| **Project Title:** | Development of an Index of Habitability Using Converging Indicators: Physiology, Performance, and Subjective Reports Physiology, Performance, and Subjective Reports | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | ADVANCED HUMAN SUPPORT TECHNOLOGIES | | |
| **Program/Discipline--Element/Subdiscipline:** | ADVANCED HUMAN SUPPORT TECHNOLOGIES--Space human factors engineering | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | Patricia.S.Cowings@nasa.gov | **Fax:** | FY 650-604-1484 |
| **PI Organization Type:** | NASA CENTER | **Phone:** | 650-604-5724 |
| **Organization Name:** | NASA Ames Research Center | | |
| **PI Address 1:** | Human Systems Integration Division (TH) | | |
| **PI Address 2:** | Mail Stop 262-2 | | |
| **PI Web Page:** | http://humansystems.arc.nasa.gov/ | | |
| **City:** | Moffett Field | **State:** | CA |
| **Zip Code:** | 94035 | **Congressional District:** | 18 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | INTERNAL PROJECT |
| **Start Date:** | 06/30/2002 | **End Date:** | 12/31/2004 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | 1 |
| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 2 | **No. of Bachelor's Degrees:** | 6 |
| **No. of Bachelor's Candidates:** | 7 | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | 0 | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| **Task Description:** | The purpose of this project is to develop and validate a quantitative method for assessing environmental effects on individual crewmembers during spaceflight. This project will involve analyses of data acquired during previous laboratory tests, field studies and spaceflight. NASA Ames Psychophysiological Research Laboratory has a unique database collected over a 25-year period on approximately 300 people exposed to a variety of environmental stressors, and the effects of both behavioral and pharmacological countermeasures on their stress tolerance. Previous research has shown that the use of converging indicators (physiological and subjective self-report measures) is a significantly more reliable method for assessing environmental effects on performance than any one indicator alone. We anticipate that the deliverable from this project will be a reliable methodology for accurately assessing microgravity effects on individual crew performance and for evaluating countermeasures designed to facilitate adaptation to space and readaptation to Earth. Transfered database to Linux platform Developed Post processing software (PostProc) Establish | | |

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| | protocols/methods for remote monitoring and training human physiological responses Data-base of biomedical and performance data from multiple experiments can be accessed from any PC platform Biomedical data can be processed, artifact removed and information derieved (e.g., cardiac output) from any PC platform Remote monitoring allows investigators from many different geographical locations to observe experiments in real-time. Remote training allows helath care providers to control physiological displays used for modifying these paramters (e.g., treatment of motion sickness, hypertension). |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | 1. Pilot (military, private. Search) training/airsickness abatement and performance enhancement.<br>2. Assessment and correction of environmental stressors effects on human health and performance.<br>3. Effective treatment for nausea and low blood pressure in various patient populations. |
| **Task Progress:** | No progress report this reporting period. Publication added to Cumulative Bibliography. |
| **Bibliography Type:** | Description: (Last Updated: 11/08/2023) |
| **Articles in Peer-reviewed Journals** | Cowings P, Toscano W, DeRoshia C, Taylor B, Hines AL, Bright A, Dodds A. "Converging indicators for assessing individual differences in adaptation to extreme environments." Aviation, space, and environmental medicine. 2007 May 1;78(5):B195-215. https://www.ingentaconnect.com/content/asma/asem/2007/00000078/a00105s1/art00029 , May-2007 |